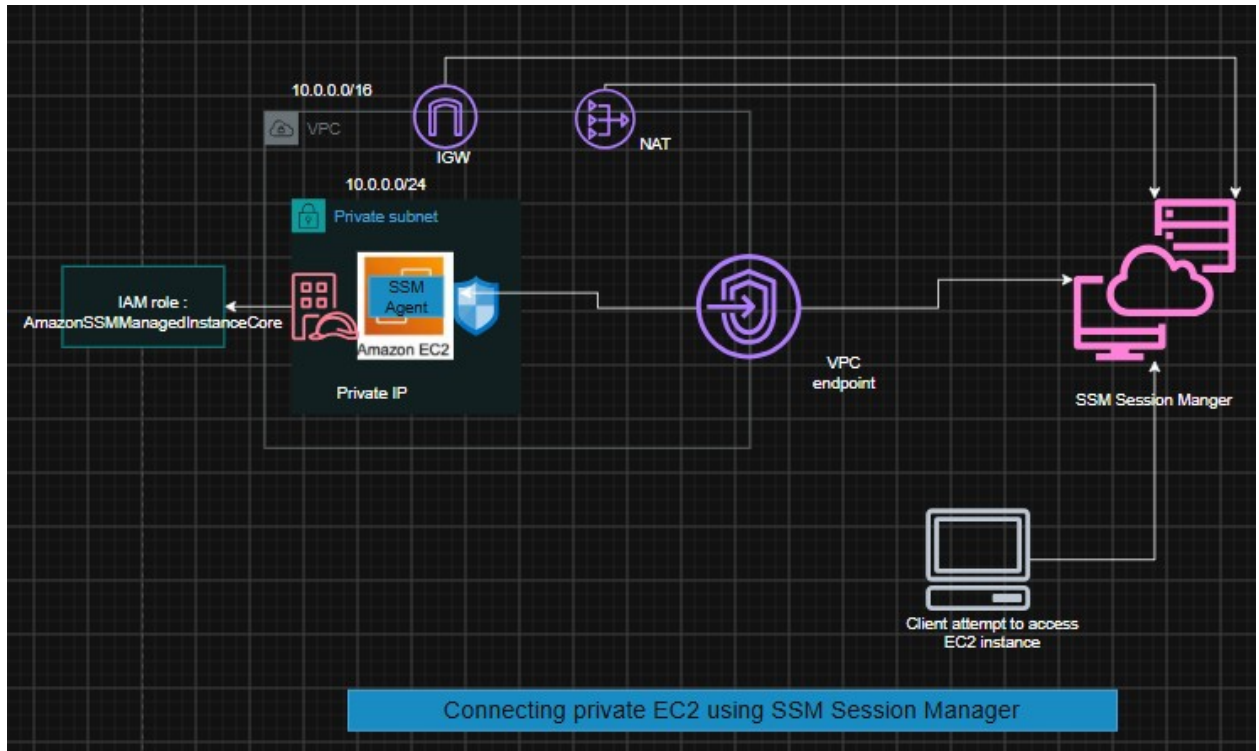


Accessing EC2 Using SSM Session Manager

1. We can connect to a public or private EC2 instance via SSM Session manager without SSH connect , STS , no need of port 22 and CLI command on AWS console.
2. We can attach IAM role (**AmozoneSSMManagedInstanceCore**) along with permission to EC2 instance to enable SSM connect .
3. EC2 instance should have **SSM agent** installed . bydfault the agent will be installed to server once you attach the role before server creation or we can configure after server creation.
4. In SG **https (443)** should be enabled.
5. SSM Session Manager service is not comes under VPC . This is outside of service.
6. We can use this service along with any one of the given service IGW , NAT and VPC end point.
7. We can also configure SSM in our local system . mind the flow.
8. Request by client -
Client request for connection -> SSM Session Manager ->IGW/NAT/VPC endpoint -> Subnet -> **EC2 instance outbound rule (SG)** ->EC2 instance .
9. Response from Ec2 instance –
EC2 instance-> **EC2 instance outbound rule (SG)**-> IGW/NAT/VPC endpoint-> SSM Session Manager-> Client system

Task -1 (We can perform this task with IG/NAT/endpoint .but here we implementing with IGW)

- Create a server and connect the server on AWS console using SSM Session Manager . (**here we are attaching IAM role after server creation**).



Prerequisite-

1. EC2 Instance
2. SG with Http port enabled
3. IAM role including (AmazonSSMManagedInstanceCore) permission.

Steps-

1. Create an instance. [Make sure attached SG, Outbound rule should include **https (443 port)**]
2. Create IAM role and add permission (AmazonSSMManagedInstanceCore).
3. Attach the permission to EC2 instance.
4. Try to connect the server

Implementation-

1.create a server name Test-SSM-Connect

Make sure enabling public ip

And do not choose any key

▼ Key pair (login) Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the inst

Key pair name - required

Proceed without a key pair (Not recommended) Default value ▼ [Create new key pair](#)

▼ Network settings Info

VPC - required Info

vpc-03bb8ae9ac6105462 (PVC-A) 10.0.0.0/16 [Create new VPC](#)

Subnet Info

subnet-03faa908ca8b3df2c Public-Subnet [Create new subnet](#)

VPC: vpc-03bb8ae9ac6105462 Owner: 198302589656 Availability Zone: us-east-1a (use 1-az1) Zone type: Availability Zone IP addresses available: 250 CIDR: 10.0.0.0/24

Auto-assign public IP Info

Enable

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group ☒ Select existing security group

Common security groups Info

Select security groups [Compare security group rule](#)

SG sg-0764de094c9c9323c X VPC: vpc-03bb8ae9ac6105462

Security groups that you add or remove here will be added to or removed from all your network interfaces.

Note - We can enable or disable public ip since we have not given any key . it wont help in connecting server with the help of SSH Clint

Server created

Instances (2) [Info](#) Last updated less than a minute ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

Saved filter sets [Choose filter set](#)

☐	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IP
☐	Flipkart	i-0ac8f89a7dd6b7947	Stopped	t3.micro	-	us-east-1a	-	-
☐	Test-SSM-Connect	i-00812d0dc54b66014	Running	t3.micro	Initializing	us-east-1a	-	44.220.1

We should get this output

EC2 Instance Connect | SSM Session Manager | **SSH client** | EC2 serial console

No associated key pair
This instance is not associated with a key pair. Without a key pair, you can't connect to the instance through SSH. You can connect using EC2 Instance Connect with just a valid username. You can connect using Session Manager if you have been granted the necessary permissions.

Instance ID
[i-00812d0dc54b66014](#) (Test-SSM-Connect)

1. Open an SSH client.
2. Locate your private key file. The key used to launch this instance is `id_rsa`.
3. Run this command, if necessary, to ensure your key is not publicly viewable.
`chmod 400 "id_rsa"`
4. Connect to your instance using its Public IP:
`44.220.167.30`

Example:
`ssh -i "id_rsa" ec2-user@44.220.167.30`

Note: In most cases, the guessed username is correct. However, read your AMI usage instructions to check if the AMI owner has changed the default AMI username.

2.Now Create IAM role and add permission(AmozoneSSMManagedInstanceCore) .

[IAM](#) > [Roles](#) > Create role

Step 1 **Select trusted entity**
Step 2 Add permissions
Step 3 Name, review, and create

Select trusted entity [Info](#)

Trusted entity type

- ☒ **AWS service**
Allow AWS services like EC2, Lambda, or others to perform actions in this account.
- ☐ **AWS account**
Allow entities in other AWS accounts belonging to you or a 3rd party to perform actions in this account.
- ☐ **Web identity**
Allows users federated by the specified external web identity provider to assume this role to perform actions in this account.
- ☐ **SAML 2.0 federation**
Allow users federated with SAML 2.0 from a corporate directory to perform actions in this account.
- ☐ **Custom trust policy**
Create a custom trust policy to enable others to perform actions in this account.

Use case
Allow an AWS service like EC2, Lambda, or others to perform actions in this account.

Service or use case
EC2

Choose a use case for the specified service.
..

Select Permission - **AmozoneSSMManagedInstanceCore**

Step 1: Select trusted entity
Step 2: **Add permissions**
Step 3: Name, review, and create

Add permissions Info

☒ Use existing policy
Attach an existing managed policy to the role.

☐ Create inline policy
Create a new policy that will be attached directly to the role.

Permissions policies (1/1190) Info

Choose one or more policies to attach to your new role.

Filter by Type: All types 37 matches

	Policy name	Type	Description
☐	AmazonEC2RoleforSSM	AWS managed	This policy will soon be deprecated. Please use AmazonSSMManagedInstanceCore pol...
☐	AmazonSageMakerCanvasSMDDataScienceA...	AWS managed	Provides permissions for Amazon SageMaker Canvas to use the SageMaker Data Scien...
☐	AmazonSSMAutomationApproverAccess	AWS managed	Provides access to view automation executions and send approval decisions to autom...
☐	AmazonSSMAutomationRole	AWS managed	Provides permissions for EC2 Automation service to execute activities defined within ...
☐	AmazonSSMDirectoryServiceAccess	AWS managed	This policy allows SSM Agent to access Directory Service on behalf of the customer fo...
☐	AmazonSSMFullAccess	AWS managed	Provides full access to Amazon SSM.
☐	AmazonSSMMaintenanceWindowRole	AWS managed	Service Role to be used for EC2 Maintenance Window
☐	AmazonSSMManagedEC2InstanceDefaultP...	AWS managed	This policy enables AWS Systems Manager functionality on EC2 instances.
☒	AmazonSSMManagedInstanceCore	AWS managed	The policy for Amazon EC2 Role to enable AWS Systems Manager service core functio...

IAM > Roles > Create role

Step 1: Select trusted entity
Step 2: Add permissions
Step 3: **Name, review, and create**

Name, review, and create

Role details

Role name
Enter a meaningful name to identify this role.
enable-ssm-session-manager
Maximum 64 characters. Use alphanumeric and '+', '@', '-', '_' characters.

Description
Add a short explanation for this role.
Allows EC2 instances to call AWS services on your behalf.
Maximum 1000 characters. Use letters (A-Z and a-z), numbers (0-9), tabs, new lines, or any of the following characters: _ + , @ - / [] ! : # \$ % ^ & * { } ~ ' " =

Step 1: Select trusted entities

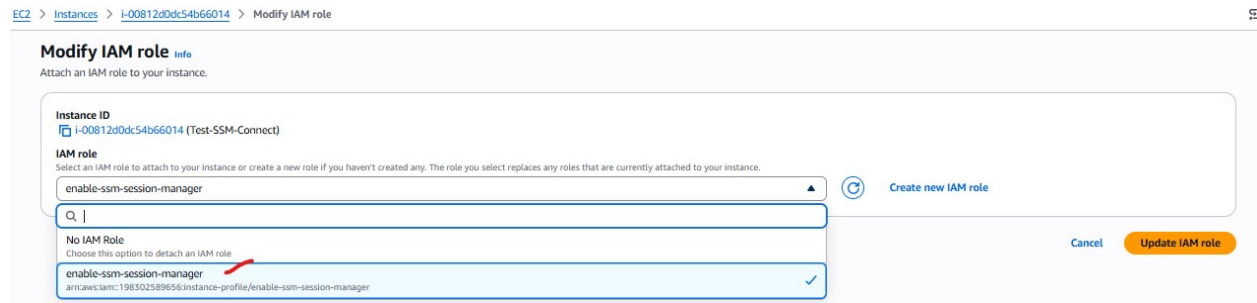
3. Attach the **enable-ssm-session-manager** role to EC2 instance (Test-SSM-Connect).

Instances (1/2) Info

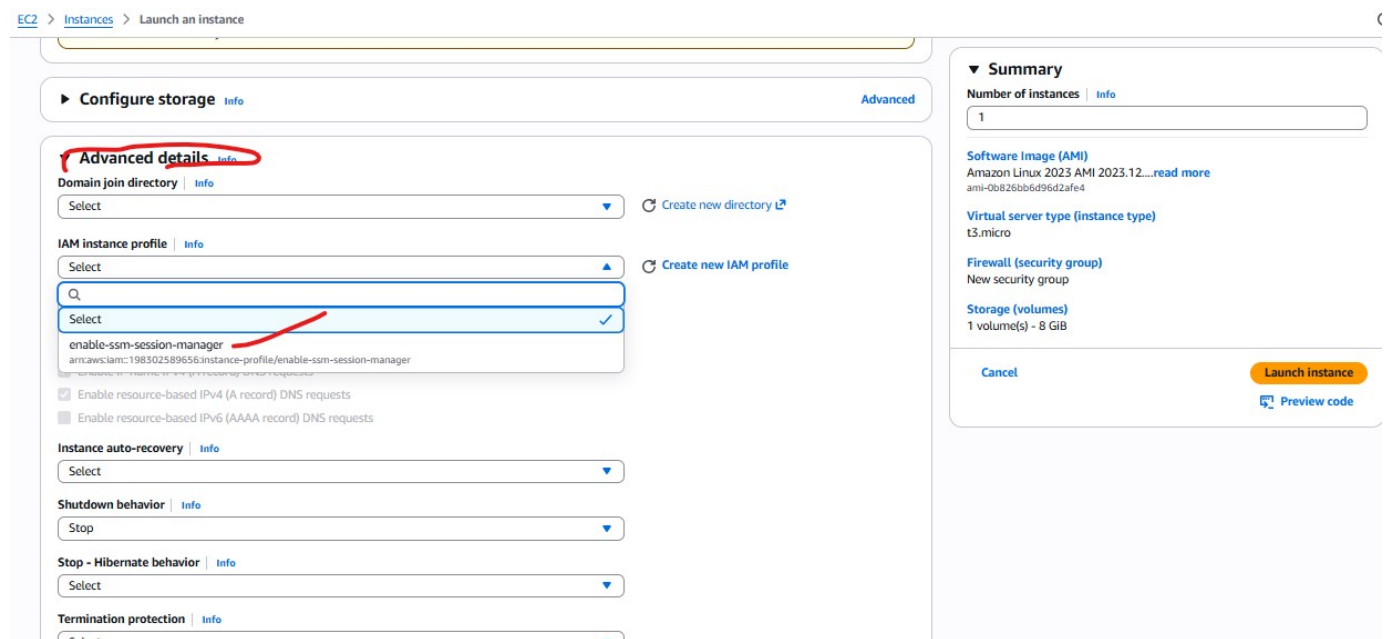
Save filter sets
Choose filter set

Find instance by attribute or tag (case-sensitive)

	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4	Actions
☐	Flipkart	i-0ac8f89a7dd6b7947	Stopped	t3.micro	3/3 checks passed	us-east-1a		Connect Instance state Instance diagnostics Instance settings Networking Security Image and templates Storage Monitor and troubleshoot
☒	Test-SSM-Connect	i-00812d0dc54b66014	Running	t3.micro	3/3 checks passed	us-east-1a		Change security groups Modify IAM role Get Windows password



*Note – We can attach the IAM role during the server creation steps.



4. Try to connect the server

It will take some time to configure then only will be able to show the connect option.

EC2 > Instances > i-00812d0dc54b66014 > Connect to instance

Connect Info

Connect to an instance using the browser-based client.

Instance ID i-00812d0dc54b66014 (Test-SSM-Connect)	VPC ID vpc-03bb8ae9ac6105462	Security groups sg-0764de094c9c9323c (SG)	IAM role enable-ssm-session-manager
--	--	---	---

EC2 Instance Connect **SSM Session Manager** SSH client EC2 serial console

SSM agent status

Ping status Online	Last ping time Fri Jul 24 2026 12:02:27 GMT+0530 (India Standard Time)	Session Manager connection status Connected	Agent version 3.3.4624.0
------------------------------	--	---	------------------------------------

Session Manager usage:

- Connect to your instance without SSH keys, a bastion host, or opening any inbound ports.
- Sessions are secured using an AWS Key Management Service key.
- You can log session commands and details in an Amazon S3 bucket or CloudWatch Logs log group.
- Configure sessions on the Session Manager [Preferences](#) page.

[Cancel](#) [Connect](#)

It will show a terminal window , there you need to give command **sudo su -** to enter the root directory.

Session ID: root-e3pcle2b3rcdg6xn87uszgjfna [Shortcuts](#) Instance ID: i-00812d0dc54b66014 (Test-SSM-Connect) [Terminate session](#)

```
sh-5.2$ sudo su -  
[root@ip-10-0-0-187 ~]#
```

Session ID: root-e3pcle2b3rcdg6xn87uszgjfna [Shortcuts](#)

```
sh-5.2$ sudo su -  
[root@ip-10-0-0-187 ~]#
```

Like this way we can connect with server using IGW(with public/pvt ip) , NAT ,custom VPC endpoint.

Note*- while Creating NAT need to select **Regional Level**.

We can configure Session manager in our local machine using following steps –

1. First configure **aws CLI** with **aws access key** and **access secret key**.
2. Use **aws configure** command.

```
➔ winget install Amazon.SessionManagerPlugin
```

Note* - sometime even after installation the PATH is not set in Environment Variable . We need to check installation by -> session-manager-plugin
And then get the path > Get-Command session-manager-plugin
Then add the path to environment

```
aws ssm describe-instance-information
```

```
aws ssm start-session --target i-086b536e26738c02b
```

```
root@ip-10-0-0-187:~
Try the new cross-platform PowerShell https://aka.ms/pscore6

. : File C:\Users\its_mukesh\Documents\WindowsPowerShell\Microsoft.PowerShell_profile.ps1 cannot be loaded because running scripts is disabled on this system. For more information, see about_Execution_Policies at https://go.microsoft.com/fwlink/?LinkID=135170.
At line:1 char:3
+ . 'C:\Users\its_mukesh\Documents\WindowsPowerShell\Microsoft.PowerShe ...
+ ~~~~~
+ CategoryInfo          : SecurityError: (:) [], PSSecurityException
+ FullyQualifiedErrorId : UnauthorizedAccess

PS C:\Users\its_mukesh> session-manager-plugin

The Session Manager plugin was installed successfully. Use the AWS CLI to start a session.

PS C:\Users\its_mukesh>
PS C:\Users\its_mukesh>
PS C:\Users\its_mukesh> aws ssm describe-instance-information
{
  "InstanceInformationList": [
    {
      "InstanceId": "i-00812d0dc54b66014",
      "PingStatus": "Online",
      "LastPingDateTime": "2026-07-24T12:57:11.986000+05:30",
      "AgentVersion": "3.3.4624.0",
      "IsLatestVersion": false,
      "PlatformType": "Linux",
      "PlatformName": "Amazon Linux",
      "PlatformVersion": "2023",
      "ResourceType": "EC2Instance",
      "IPAddress": "10.0.0.187",
      "ComputerName": "ip-10-0-0-187.ec2.internal",
      "SourceId": "i-00812d0dc54b66014",
      "SourceType": "AWS::EC2::Instance"
    }
  ]
}

PS C:\Users\its_mukesh> aws ssm start-session --target i-00812d0dc54b66014

Starting session with SessionId: root-uhbltx54p8nn29qpygs2iavsa
sh-5.2$ sudo su -
Last login: Fri Jul 24 07:01:59 UTC 2026 on pts/1
[root@ip-10-0-0-187 ~]# ls
Hi.txt
[root@ip-10-0-0-187 ~]#
```

Task Over